

NebAI: A Desktop AI Assistant

Project Report

Objective

The goal of this project is to build a simple desktop AI assistant using Python. The assistant helps the user with everyday tasks like checking the time, taking notes, and setting reminders, using both voice commands and a simple button-based interface.

What It Does

- Tells the current time and day.
- Listens to voice commands and responds accordingly.
- Takes notes by voice and saves them with a timestamp.
- Reads saved notes back out loud.
- Sets reminders and alerts the user when they're due.
- Speaks back to the user using text-to-speech.

How It Was Built

The project is written in Python. It uses **Tkinter** for the graphical interface, **pyttsx3** for text-to-speech, and **SpeechRecognition** to convert the user's voice into text commands. When the user speaks or clicks a button, the assistant matches the command to a function (like telling the time or taking a note) and responds with both an on-screen message and a spoken reply. Notes are saved to a text file, and reminders run in the background using a separate thread so the app stays responsive.

Results

The assistant works as expected. It correctly recognizes voice commands, tells the time and day, saves and reads back notes, and triggers reminders at the right time. Errors, such as not hearing the user clearly, are handled gracefully instead of crashing the app.

Conclusion

This project shows how different Python tools can be combined to build a working, voice-enabled assistant. It successfully completes its main goals and could be extended in the future with more commands and smarter conversation handling.